

Colin Ray Tarkington

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Summary

Detail-oriented cybersecurity and information security student with hands-on research and industry experience at Oak Ridge National Laboratory and Shaw Industries. Experienced in vulnerability research, embedded systems and IoT security, security monitoring, and threat detection. At Shaw Industries, gained practical experience developing KQL-based detection rules and alerts in Microsoft Sentinel, working with Microsoft Entra ID, and designing an isolated Kali Linux sandbox environment to support purple-team security testing. Proven ability to conduct vulnerability assessments, analyze network traffic, simulate real-world attack scenarios, and translate technical findings into actionable security improvements. Strong foundation in computer science and cybersecurity, complemented by effective communication, teamwork, and professional experience. Seeking opportunities to contribute to security operations, threat detection, vulnerability research, and secure system design.

EDUCATION

University of Tennessee, Knoxville, Anticipated June 2027 • B.S. Information/Cyber Security

- Cumulative GPA: 3.8
- Relevant Coursework: Introduction to Cyber Security, Fundamentals of Information Technology, Programming for Information Science (Python), Information Management, Public Relations, Database Design, Cyber Risk Management

Pellissippi State Community College, Graduated June 2025 • A.S. Computer Science

- Cumulative GPA: 3.5
- Relevant Coursework: Fundamentals of Communication, Computer Science 1, Computer Science 2, Assembly and Computer Organization, Calculus 1, Calculus 2

EMPLOYMENT EXPERIENCE

Intern: Cyber Operations, June 2026-August 2026, Shawn Industries – Dalton, GA

- Developed and implemented Microsoft Sentinel detection engineering use cases by researching adversary techniques, mapping behaviors to the MITRE ATT&CK framework, and creating KQL-based detections focused on identity attacks, MFA abuse, privilege escalation, suspicious PowerShell activity, inbox rule manipulation, and account takeover scenarios.
- Conducted threat modeling and cybersecurity research on emerging attack techniques, including analysis of payroll fraud and account takeover campaigns, to identify detection opportunities and strengthen organizational visibility into high-risk activity.
- Designed and enhanced server monitoring and logging health solutions within Microsoft Sentinel by developing KQL analytics to identify heartbeat failures, log ingestion outages, and visibility gaps across enterprise server environments.

- Collaborated with senior cybersecurity engineers to improve detection logic, reduce false positives, and implement threshold-based alerting methodologies for operational monitoring and incident response workflows.
- Architected a Kali Linux Purple Team testing environment for detection validation and controlled security testing, including hardware requirements, network segmentation, firewall policies, credential governance, access controls, logging requirements, and security monitoring considerations.
- Authored Enterprise Architecture review documentation, technical proposals, architecture diagrams, and executive presentations to support organizational approval and governance of a dedicated cybersecurity testing platform.

Intern: Embedded Systems Research, June 2025–Dec 2025, Oak Ridge National Laboratory—Oak Ridge, TN

- Analyzed and monitored over-the-air (OTA) data transmissions to detect anomalies and potential attack vectors.
- Conducted advanced vulnerability research on IoT devices to identify security weaknesses in smart home ecosystems.
- Developed and executed automated scripts simulating real-world user interactions to evaluate device resilience under realistic conditions.
- Learned how to use cybersecurity tools to advance research.
- Utilized tools to create attack scenarios on specific systems for research purposes.
- Performed data-driven assessments to uncover exploitable flaws and improve device security posture.
- Collaborated with researchers to propose mitigation strategies and enhance IoT security standards.

Intern: Embedded Systems Research, June 2024 – Aug 2024, Oak Ridge National Laboratory, Oak Ridge, TN

- Developed skills in soldering and put together an ultrasonic levitator.
- Trained on a rework station to remove individual chips from circuit boards.
- Conducted vulnerability assessments on smart home devices and systems.
- Learned how to use cybersecurity tools to advance research.
- Utilized tools to create attack scenarios on specific systems for research purposes.
- Wrote and published an in-depth research paper on work conducted.
- Built a poster presentation that explored the results of research.

Part-time Sales Associate, May 2022 - August 2023, September 2024 - Current: Nike, Knoxville, TN

- Organized racks and shelves to maintain store visual appeal, engage customers, and promote specific merchandise.
- Helped customers locate products and checked the store system for merchandise at other locations.
- Prepared merchandise for sales by pricing or tagging.
- Rotated stock and restocked shelves to maintain product availability and store appearance.
- Managed returns, exchanges, and refunds in accordance with store policy.
- Answered customer questions about sizing, accessories, and merchandise.
- Maintained a calm demeanor and professionally managed issues in busy, high-stress situations.
- Engaged with customers to build relationships and loyalty.

Inventory Associate Feb 2024–June 2024 CarMax-Knoxville, TN

- Responsible for the transporting and receiving process of vehicles.
- Prepared vehicles for customer delivery.
- Merchandised the lot to create an accessible and easy experience for customers.

- Performed visual examinations of automobiles, making notes about the modifications required to turn them into cars that surpass expectations.
- Maintained the cars' overall condition by replacing and repairing tiny plastic and other parts.
- Retouched paint, finished paint, waxed, and buffed cars to restore their original appearance.

Fitness Floor Staff Member, August 2021-May 2022, Davis Family YMCA, Knoxville, TN

- Responsible for helping members with basic use and operation of fitness equipment.
- Stayed available on the exercise floors for basic information and advice.
- Keep the floor clean and neat
- Developed positive relationships with members and provided motivational support and guidelines.
- Increased participants' awareness of all healthy lifestyle factors.
- Ensure safe and effective workouts.
- Keep certifications, training, and knowledge current.

EXTRACURRICULAR ACTIVITIES & LEADERSHIP POSITIONS

Dean's List Fall 2022-Spring 2024, Pellissippi State Community College

- A scholarly award for students who demonstrate academic excellence.

Inducted Member of the National Society of Leadership and Success, Fall 2022, Pellissippi State Community College

- A community of goal-oriented individuals seeking professional leadership training, exclusive scholarships, and networking events.

TECHNICAL SKILLS

- Programming Languages: C/C++, Python, HTML, CSS
- Security Tools: Aircrack-ng, Wireshark, Aircrack-ng, Bettercap, Nmap, Metasploit, Microsoft Sentinel • KQL (Kusto Query Language) • Detection Engineering • Threat Modeling • MITRE ATT&CK • Microsoft Defender
- Microsoft Entra ID • SIEM Operations •

References available upon request